

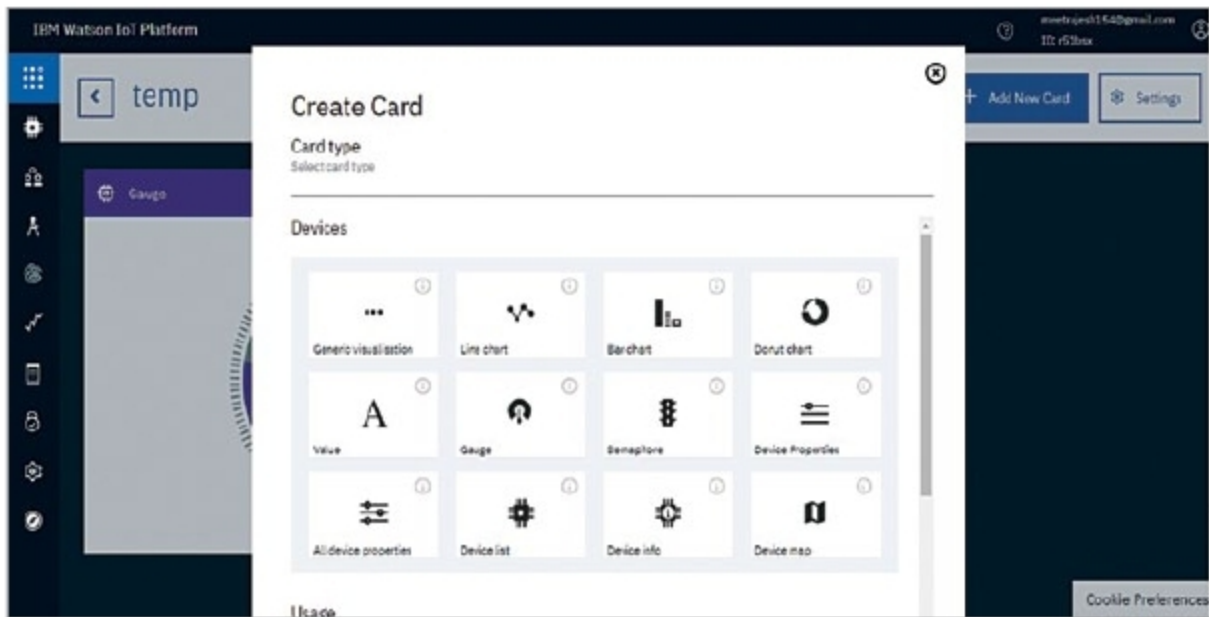


Fig. 16: Various card options

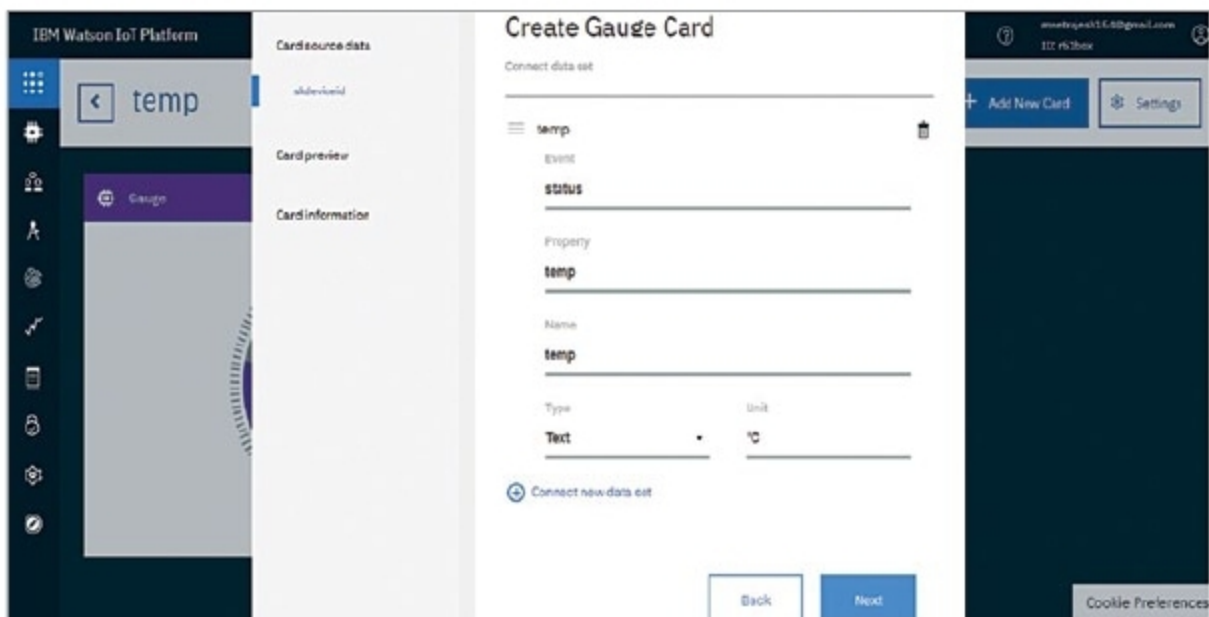


Fig. 17: Create Gauge Card

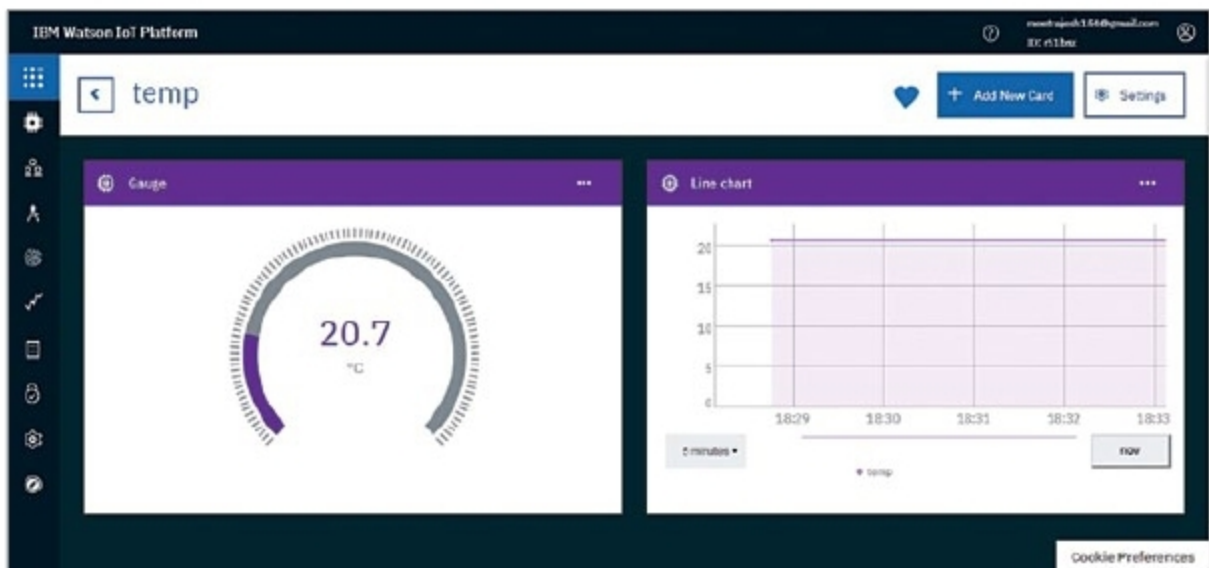


Fig. 18: Screen showing sensor data on gauge and line chart

shown in Fig. 10. Click on it.

Now, create Device Type and Device ID, as shown in Fig. 11, and click on Next.

Do not do anything on the next screen, just click on Next.

A token will be generated and added, as shown in Fig. 12. Note down these details because you will need the token, device ID and device type details in Python program later.

Connection status will be as shown in Fig. 13.

Remember to change security level to TLS Optional, otherwise it will not allow you to publish.

Open Python code and enter your device ID, type and authentication token. A typical example of Python program with device ID is shown in Fig. 14.

Next, run Python program to read and publish temperature data to the cloud from the PC connected to the temperature meter.

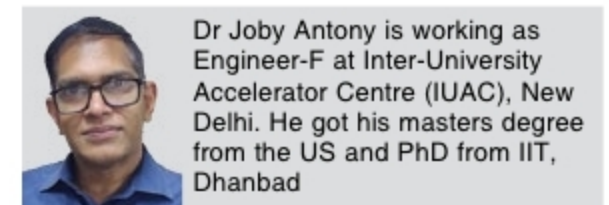
Data can be displayed on a line chart, bar graph, gauge and so on. You can view the data better on the screen using any one of these options. These options can be added using Card on IBM Bluemix cloud. In our case, a gauge was selected for viewing online temperature.

First, open IBM cloud, click on Boards to create a board, as shown in Fig. 15.

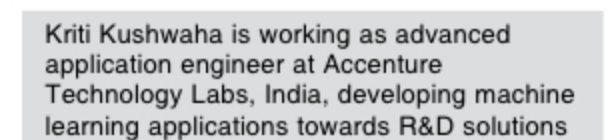
Then create a card on Create Card window, as shown in Fig. 16.

Next, a screen will appear with device ID (for example, Skdeviceid in this case), as shown in Fig. 17. Click on Next.

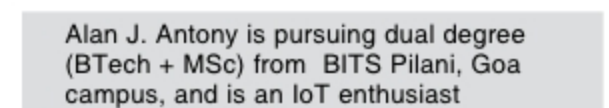
Temperature data will appear on screen that will show sensor data on gauge and line chart, as shown in Fig. 18. **EFY**



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EFY Note

The source codes and more testing details of this project are available for free download at source.efymag.com

Click on Connections on the left. Then click on connection name (Sk1234-iotf-service in this case).

Next, click on Internet of Things Platform. A window will appear as shown in Fig. 9.

Click on Manage on the next screen. You will see a Launch button. Click on it and you will see Add Device button at the top-right corner, as